

Ramiro Barrantes Reynolds
CURRICULUM VITAE

Position: Bioinformatics Scientist
Department of Microbiology and Molecular Genetics

Address: Stafford Hall 220A
Department of Microbiology and Molecular Genetics
The Robert Larner, M.D. College of Medicine
University of Vermont
Burlington, VT 05405
email: ramiro.barrantes@med.uvm.edu

EDUCATION

Year	Institution	Degree	Area of Degree, Accolades
2011-2011	University of Vermont (Brian Beckage)	Postdoctoral	Bayesian Statistical methods in Ecology
2006-2011	University of Vermont	Ph. D.	Cell and Molecular Biology with a focus on Bioinformatics/Statistics
1994-1999	University of Costa Rica	B.S.	Pure Mathematics
1994-1999	University of Costa Rica	B.S.	Computer Science

LICENSES, CERTIFICATION

Years	License/Certification
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FACULTY POSITIONS HELD

Years	Institution	Academic Title	Department
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OTHER POSITIONS AND MAJOR ADMINISTRATIVE POSITIONS HELD

Years	Location/Program Name	Role
2022-present	UVM Vermont Integrative Genomics Resource	Bioinformatics Scientist
2022-present	UVM Translational Global Infectious Disease Research	Bioinformatics Scientist
2011-2021 (part time)	UVM Bioinformatics Shared Resources	Bioinformatics Scientist
2011-2021	Precision Bioassay	Statistician
2001-2006	University of Vermont	Bioinformatics Programmer

HONORS AND AWARDS – none to date

Year	Name of Award
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KEYWORDS/AREAS OF INTEREST *Please provide a set of MESH indexing terms to describe your research and/or clinical interests./*

Bioinformatics, Data Science, Genomics, Epigenomics, Variant Calling, Bayesian Analysis, Machine Learning, Artificial Intelligence, Data Integration, Reproducibility, Workflow Management Systems, Reproducibility

SUMMARY OF PROFESSIONAL ACTIVITIES- OVERALL

Since 2001, my career has focused on integrating quantitative sciences, computational methods, and biomedical research through collaborative partnerships with scientists. Since joining UVM full-time in July 2022, I have collaborated on short- and long-term projects with 12-15 UVM investigators, two core facilities (Bioinformatics Shared Resource and the Genomics facility), contributing to five grant submissions, five publications (one accepted, two submitted, two in preparation), five open-source software modules, 2-3 operational bioinformatics pipelines for core facilities, and one bioinformatics web platform.

My recent scientific work at UVM encompasses bioinformatics and computational biology methods spanning variant analysis, epigenomics, virus typing, transcriptomics, B-cell and T-cell receptor profiling, and structural biology. The quantitative scope ranges from sample size calculations and implementation of statistical methods to sophisticated probabilistic inference and machine learning applications. My earlier work included extensive experience in sequence analysis, phylogenetics, and DNA repair enzyme modeling, utilizing mixed models, resampling techniques, and machine learning approaches.
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I work extensively with workflow management systems, containerized environments, version control, and high-performance computing resources as a way to enforce reproducibility and scalability. Effective communication with collaborators is central to my approach. I have also played a key role in building reproducible computational infrastructures for core facilities, dating back to the early 2000s with the creation of BioDesktop—a platform that handled thousands of orders for the UVM DNA Analysis Facility—and more recently implementing Nextflow pipelines deployed through the Seqera platform.

Prior to UVM, I spent ten years in industry as a statistician at a statistical consulting firm (while working part time at UVM), where I gained extensive experience in experimental design and mixed nonlinear models while developing a deep appreciation for precision and rigor in client-facing work, which also included developing a software product for analysis of bioassay data.

From a teaching and mentoring perspective, I have delivered guest lectures in the statistics and bioinformatics components of MMG courses and have mentored/supervised five students through investigator collaborations. Looking forward, I am deeply interested in artificial intelligence and eager to learn and implement and apply these methods in research.

SUMMARY OF ACCOMPLISHMENTS

- Integral component as the data scientist and bioinformatician member of the Bioinformatics Shared Resource and the Translational and Infectious Disease team in collaborations with many UVM scientists, leading to multiple publications.
- Part of the lead programming team for a bioinformatics web platform: BioDesktop (UVM) and for a statistical analysis web product Xymp (Precision Bioassay Inc.)
- Bioinformatics key personnel member of the team in the development of a personalized antibody therapy for the treatment of cancers. This work started in 2022 and is ongoing.

PROFESSIONAL SERVICE

DEPARTMENTAL SERVICE

Years	Department	Committee	Role
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COLLEGE SERVICES

Years	Service Committee	Role
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MEDICAL CENTER SERVICE

Years	Service Committee	Role
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UNIVERSITY SERVICE

Years	Service Committee	Role
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GOVERNMENT

Years	Service Committee	Role
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SOCIETY MEMBERSHIPS

Years	Society
2009-present	American Statistical Association
2010-present	International Society for Bayesian Analysis
2025-present	Association of Biomolecular Resource Facilities (ABRF)
2009-2011	International Society for Computational Biology

SERVICE TO PROFESSIONAL ORGANIZATIONS

Years	Service Committee	Role
2025-present	Nextflow	Ambassador

SERVICE TO PROFESSIONAL PUBLICATIONS

Years	Journal/Publication/Board
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PUBLIC SERVICE

Years	Service Role
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2001-present	Volunteer at the Vermont Zen Center Buddhist community. Help with fundraisers, translator/interpreter, communication technology, guest master, housework supervisor, pilgrimage coordinator.
2020	Creator and host for the Phoneaholics Anonymous podcast. Resource to help people have a better relationship with technology
2001-2010	Northwest State Correctional Facility in St. Albans, Volunteer with meditation group for inmates and later maintained correspondence.
2006	Spectrum Youth and Family Services dance teacher volunteer for at-risk youth.

SUMMARY OF SERVICE ACTIVITIES

For many years I have worked on myself through meditation practices and also in the area of non-violent communication and conflict resolution. My service has mostly been around volunteering at the Buddhist community that I am part of (to this day many hours per week), which extended for many years to working as a meditation volunteer to help inmates at a prison in St. Albans. Furthermore, since I work so much in technology, I have noticed first hand the impact that it has in our attention and have tried to bring awareness to that personally and to the greater community to a podcast.

TEACHING

FORMAL SCHEDULED COURSES [list classes taught since last promotion. This is the same table and format found in the COM Teaching Academy Portfolio. Add same number of lines received from Teaching Academy Portfolio, then paste cells in.]

Year	Course Title	Course R E	Hours	Number of Learners	Learner Level
2021	MMG231: Bioinformatics & Data Analysis	E	4	25	U, G
2021	MMG232: Advanced Bioinformatics	E	4	16	U, G
2020	MMG232: Methods in Bioinformatics	R(U),E(G)	4	28	U, G
2018	MMG232: Methods in Bioinformatics	R(U),E(G)	4	25	U, G
2017	MMG232: Methods in Bioinformatics	R(U),E(G)	4	26	U, G
2016	MMG232: Methods in Bioinformatics	E	4	24	U, G
2015	MMG232: Molecular Bioinformatics	E	4		U, G

R-required; E-elective; Hours-approx. per semester; G-graduate studies (instruction as per the LCOM Teaching Academy Portfolio)

CURRICULUM DEVELOPMENT**POSTGRADUATE AND OTHER COURSES**

Years Department Responsibility

PREDOCTORAL STUDENTS SUPERVISED OR MENTORED

Dates	Name	Program School	Role	Current Position
2024-present	Kayla Liebeskind	MMG	Supervised/Mentor	Undergraduate
2024-2025	Alexis Robertson	MMG	Supervised/Mentor	Graduate school
2024-2025	Allyson Turner	MMG	Supervised/Mentor	Lab technician
2023-2024	Ava Trueworthy	Computer Science	Supervised/Mentor	Undergraduate
2024	Zachary Miller	MMG	Supervised	Graduate school

DISSERTATION/THESIS COMMITTEE MEMBERSHIP**POSTDOCTORAL FELLOWS AND RESIDENTS DIRECTLY SUPERVISED OR MENTORED**

Dates	Name	Fellow	Faculty Role	Current Position
2022-2023	Julia Snyder			Industry

INFORMAL TEACHING

Year One Sentence Description of Teaching
 2024 Introduction to running Nextflow pipelines and the Seqera Platform.

FACULTY MENTORED

Dates	Name	Position while Mentored	Faculty Role	Current Position

OTHER VISITING FACULTY SUPERVISED

Years	Names	University/College
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TEACHING AIDS

TEACHING AWARDS AND NOMINATIONS

Year	Award
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SUMMARY OF TEACHING ACTIVITIES

My teaching has been mostly focused in helping students and collaborators learn and use bioinformatics programming and methods, which I have found to be skills that are lacking in many of the collaborations and that students are eager to learn. I have had the pleasure of working and mentoring several students and I am exploring what is the best way to get them up to speed on bioinformatics techniques. I have also been guest lecturer on bioinformatics and statistics for several courses over the years.

RESEARCH AND SCHOLARLY ACTIVITIES

RESEARCH AWARDS AND GRANTS

Ongoing Research Support

P20 GM125498 Beth Kirkpatrick (PI) 08/2023-07/2028
Translational Global Infectious Disease Research (TGIR) Center (COBRE) – Phase II

Role: Bioinformatician

ACTIVE RESEARCH SUPPORT- Other

Title: Development of immunotherapy for cancer

Funding Source: SD Ireland Cancer Research Foundation

Funding Period: Renewed annually

Role: Key Personnel (Recipient- Krag)

Goal of Project: *The goal of this current research is to develop personalized antibody therapy for the treatment of cancers*

Completed Research Support

KZTCF7AFBRF1 David Lansky (PI) 03/2021-10/2022
Better bioassays via designs for robots analyses with improved model selection and similarity bounds that limit potency bias

Role: Co-Investigator

David Lansky (PI) 2017
Equivalence-based Random Effects Model Selection for Bioassay

Role: Co-Investigator

Pending

1K08CA300927-01A1 Volaric (PI) Submitted: 06/13/24 Score: (3%)
EBV-positive diffuse large B-cell lymphoma: defining biologic determinants of disease pathogenesis in immunodeficiency.

Role: Scientific
Advisor

Foundation Grant The Steven and 7/2026 – 6/2031 (Pending funding
Alexandra Cohen decision)
Foundation

The Cohen RENEWAL Center for Women in Midlife and Menopause: A Clinical Center of Excellence at the University of Vermont (MPI Kallen/ Madsen)

Goal: To create and build a clinical center of excellence with a robust associated research program targeted to improving the health of women in midlife and menopause across rural VT, NY, NH.

Role: Co-Investigator

Selected Unfunded Grant Submissions (include scoring if available)

SCHOLARSHIP

Peer Reviewed Publications

Original Research

1. Volaric, A.K.; **Barrantes-Reynolds, R.**; Sahakyan, K.; Fedoriw, Y.; Fietze, S. The EBV-Positive Tumor Methylome Is Distinct from EBV-Negative in Diffuse Large B-Cell Lymphoma. *Cancers* 2025, *17*, 2994
2. Budd RC, Scharer CD, **Barrantes-Reynolds R**, Legunn S, Fortner KA. T Cell Homeostatic Proliferation Promotes a Redox State That Drives Metabolic and Epigenetic Upregulation of Inflammatory Pathways in Lupus. *Antioxid Redox Signal*. 2022 Mar;36(7-9):410-422. doi: 10.1089/ars.2021.0078. Epub 2021 Nov 9. PMID: 34328790; PMCID: PMC8982120.

3. **Barrantes-Reynolds, R.**, Bond, J. P., Wallace S. 2010. Using Shifts in Amino Acid Frequency and Substitution Rate to Identify Latent Structural Characters in Base-Excision Repair Enzymes. PlosONE, 2011
4. Hatle, K. M., W. Neveu, O. Dienz, S. Rymarchyk, **R. Barrantes**, S. Hale, N. Farley, K. M. Lounsbury, J. P. Bond, D. Taatjes and M. Rincon (2007). "Methylation-controlled J protein promotes c-Jun degradation to prevent ABCB1 transporter expression." Mol Cell Biol 27(8): 2952-2966.
5. Jaworski, D. M., M. Beem-Miller, G. Lluri and **R. Barrantes-Reynolds** (2007). "Potential regulatory relationship between the nested gene DDC8 and its host gene tissue inhibitor of metalloproteinase-2." Physiol Genomics 28(2): 168-178.
6. Kathe, S. D., **R. Barrantes-Reynolds**, P. Jaruga, M. R. Newton, C. J. Burrows, V. Bandaru, M. Dizdaroglu, J. P. Bond and S. S. Wallace (2009). "Plant and fungal Fpg homologs are formamidopyrimidine DNA glycosylases but not 8-oxoguanine DNA glycosylases." DNA Repair (Amst) 8(5): 643-653.
7. Murphy, J. A., **R. Barrantes-Reynolds**, R. Kocherlakota, J. P. Bond and M. S. Greenblatt (2004). "The CDKN2A database: Integrating allelic variants with evolution, structure, function, and disease association." Hum Mutat 24(4): 296-304.
8. Robey-Bond, S. M., **R. Barrantes-Reynolds**, J. P. Bond, S. S. Wallace and V. Bandaru (2008). "Clostridium acetobutylicum 8-oxoguanine DNA glycosylase (Ogg) differs from eukaryotic Oggs with respect to opposite base discrimination." Biochemistry 47(29): 7626-7636.
9. Tang, G., T. Ruiz, **R. Barrantes-Reynolds** and K. P. Mintz (2007). "Molecular heterogeneity of EmaA, an oligomeric autotransporter adhesin of Aggregatibacter (Actinobacillus) actinomycetemcomitans." Microbiology 153(Pt 8): 2447-2457.

In Review–

1. Rab11B is required for binding and entry of recent H3N2, but not H1N1, influenza A isolates". Allyson Turner, Sara Jaffrani, Hannah Kubinski, Deborah Ajayi, Matthew Owens, Conor Fanuele, Madeline McTigue, Cailey Appenzeller, Addington Bowling, Hannah Despres, Madaline Schmidt, David Shirley, Jessica Crothers, **Ramiro Barrantes-Reynolds**, and Emily Bruce. Submitted to Journal of Virology
2. Tumor-specific antibody cocktail treatment suppresses colorectal tumor growth in mice. Krag, D., Shukla G., Pero S., Yujing S., Mei L., **Barrantes-Reynolds R.**, Fournier M., Ackerman M. Submitted to Cancer Immunology

In Preparation

1. Bayesian Pathogenic Variants Prediction using Structural Information. Liebeskind K., Barrantes-Reynolds R., Francklyn C.

Non-Peer Reviewed Publications

Review Articles

Books and Chapters–

1. Pumo DE, **Barrantes-Reynolds R**, Kathe SD, Wallace SS, Bond JP. In: Genome Instability and Transgenerational Effects. Kovalchuk I, Kovalchuk O, editors. Nova Science Publishers, Inc.; New York: 2010. pp. 71–88.

Other Scholarly Publications

2. Monika Joshi, Stephanie C Pero, Yves Durocher, Yujing Sun, Girja Shukla, Colleen Murphy, **Ramiro Barrantes-Reynolds**, Catherine Bernard, Gary Welch, David N. Krag; Penn State Cancer Institute, Hershey, PA; University of Vermont Larner College of Medicine, Department of Surgery, Burlington, VT; National Research Council Canada, Montreal, QC; Kaleidoscope Innovation, Cincinnati, OH; University of Vermont Integrative Genomics Resource, Burlington, VT; International Regulatory Affairs Services Inc., Denver, CO; Snowpeak Consulting, Truckee, CA; University of Vermont Larner College of Medicine, Department of Surgery and Moonshot Antibodies, Inc., Burlington, VT. 2026. Phase 1, open-label clinical trial to treat stage IV cancer patients harboring multiple patient-specific mutated cell surface proteins with chimeric antibodies. American Society of Clinical Oncology Meeting. May 29-June 2, 2026
3. **Barrantes-Reynolds, R.**, Bond, J. P., Wallace S. 2011. Bayesian Model to Identify Groups of Amino Acids that Change in Selection Pressure through Evolution. Poster at the Workshop on Computational Statistical Methods for Genomics and Systems Biology. April 18-22, 2011
4. **Barrantes-Reynolds, R.**, Bond, J. P. 2010. Using shifts in amino acid frequency or substitution rate to identify latent structural characters improves our understanding of the structure, function and evolution of base-excision repair enzymes. Poster at the Ninth Valencia International Meeting on Bayesian Statistics. ISBA, June 3-8, 2010.

Patents Issued or Pending

Other Creative Activities

1. Maintainer for vigr website (2022-2023)
2. Choreographer and creator for Dance your PhD thesis representation as dance: <https://vimeo.com/14608975>
3. Latin percussionist and singer for Los Sónoros Cuban/Brazilian band

Quality Improvement and Patient Safety Activities

SUMMARY OF SCHOLARLY ACTIVITIES

My work has involved applying existing bioinformatics methods and technologies, and applying or developing quantitative methods for research in Bayesian and probabilistic modeling of protein and enzyme evolution, DNA repair and redox biology, and applied genomics/bioinformatics for high-throughput and viral sequencing data, and more recently cancer epigenomics and infectious disease immunology. This has led to multiple publications, with more in preparation. While working in industry, where publishing was not the focus, I did research on non-clinical statistics methods used internally to inform the work with clients.

INVITED PRESENTATIONS *List all as appropriate, at least 5 years.*

Regional

Years

Host Organization

City, State

1/21/2026

02/13/2025

University of Vermont
Bioinformatics Workflows with Nextflow and NF-
Core

Ramiro Barrantes-Reynolds
Burlington, Vermont

10/18/2023

NERLSCD
Reproducibility in Bioinformatics

Burlington, Vermont

National

04/19/2025

Association of Biomolecular Resource Facilities
Managing Data Analyses with Nextflow and nf-core
in Bioinformatics Core Facilities

Las Vegas, Nevada

International

03/28/2005

Kyoto University
UVM Biodesktop: A flexible, secure software
system that permits data,
database, and user profile maintenance for
biological researchers

Kyoto, Japan